RELATIVE PSEUDO-COHERENCE IS A TWO-OF-THREE PROPERTY ON DISTINGUISHED TRIANGLES, WITH THE SHARP M/M+1 BOOKKEEPING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17015-17032 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17034-17037, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Author's own typo kept verbatim in the statement: 'the so is the third' (lines 17030); not corrected, per lane policy (L16119 precedent).

Editorial note. Sister card to this wave's L16191/L16194/L16195: together the four A-side relative-pseudo-coherence formal lemmas.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma relative-pseudo-coherence-characterize (S9 registered); Cohomology, lemma cohomology-lemma-cone-pseudo-coherent.

Source reference. Original lemma 08CD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17037–17037. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09UJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16837–16847. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let (E, E', E'') be a distinguished triangle of $D(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$.*

1. *If E is $(m+1)$-pseudo-coherent relative to S and E' is m-pseudo-coherent relative to S then E'' is m-pseudo-coherent relative to S.*
2. *If E, E'' are m-pseudo-coherent relative to S, then E' is m-pseudo-coherent relative to S.*
3. *If E' is $(m+1)$-pseudo-coherent relative to S and E'' is m-pseudo-coherent relative to S, then E is $(m+1)$-pseudo-coherent relative to S.*

Moreover, if two out of three of E, E', E'' are pseudo-coherent relative to S, the so is the third.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Immediate from Lemma 09UJ and Cohomology, Lemma 08CD. □